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TOUCH ELECTRODE ARCHITECTURE FOR HIGH-TRANSMITTANCE TOUCH SCREEN

Abstract

Touch electrode architecture techniques can be used to reduce or eliminate metal mesh within the one or more high-transmittance regions of a touch screen including one or more high-transmittance regions. In some examples, one or more optical devices can be integrated with a touch screen such that light associated with the one or more optical devices passes through one or more layers of the touch screen. In some such examples, to avoid degrading performance of the optical devices, one or more high-transmittance regions can be used. Additionally or alternatively, in some examples, the high-transmittance can be achieved using touch electrode architecture techniques that use transparent or semi-transparent materials instead of opaque metal mesh within the high-transmittance regions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/174,425, filed Feb. 24, 2023, published on Sep. 21, 2023 as U.S. Publication No. 2023-0297199, which claims the benefit of U.S. Provisional Application No. 63/268,754, filed Mar. 1, 2022, the contents of which are herein incorporated by reference in their entireties for all purposes.

FIELD OF THE DISCLOSURE

[0002] This relates generally to touch sensor panels, and more particularly to touch electrode architectures for high-transmittance touch screens or touch screens including high-transmittance regions.

BACKGROUND OF THE DISCLOSURE

[0003] Many types of input devices are presently available for performing operations in a computing system, such as buttons or keys, mice, trackballs, joysticks, touch sensor panels, touch screens and the like. Touch screens, in particular, are popular because of their ease and versatility of operation as well as their declining price. Touch screens can include a touch sensor panel, which can be a clear panel with a touch-sensitive surface, and a display device such as a liquid crystal display (LCD), light emitting diode (LED) display or organic light emitting diode (OLED) display that can be positioned partially or fully behind the panel so that the touch-sensitive surface can cover at least a portion of the viewable area of the display device. Touch screens can allow a user to perform various functions by touching the touch sensor panel using a finger, stylus or other object at a location often dictated by a user interface (UI) being displayed by the display device. In general, touch screens can recognize a touch and the position of the touch on the touch sensor panel, and the computing system can then interpret the touch in accordance with the display appearing at the time of the touch, and thereafter can perform one or more actions based on the touch. In the case of some touch sensing systems, a physical touch on the display is not needed to detect a touch. For example, in some capacitive-type touch sensing systems, fringing electrical fields used to detect touch can extend beyond the surface of the display, and objects approaching near the surface may be detected near the surface without actually touching the surface.

[0004] Capacitive touch sensor panels can be formed by a matrix of partially or fully transparent or non-transparent conductive plates (e.g., touch electrodes) made of materials such as Indium Tin Oxide (ITO). In some examples, the conductive plates can be formed from other materials including conductive polymers, metal mesh, graphene, nanowires (e.g., silver nanowires) or nanotubes (e.g., carbon nanotubes). It is due in part to their substantial transparency that some capacitive touch sensor panels can be overlaid on a display to form a touch screen, as described above. Some touch screens can be formed by at least partially integrating touch sensing circuitry into a display pixel stack-up (i.e., the stacked material layers forming the display pixels).

BRIEF SUMMARY OF THE DISCLOSURE

[0005] This relates generally to touch sensor panels, and more particularly to touch electrode architectures for high-transmittance touch screens or touch screens including high-transmittance regions. In some examples, one or more optical devices can be integrated with a touch screen such

that light associated with the one or more optical devices passes through one or more layers of the touch screen. In some such examples, to avoid degrading performance of the optical devices, one or more high-transmittance regions can be used. In some examples, the high-transmittance can be achieved using touch electrode architecture techniques that reduce or eliminate metal mesh within the high-transmittance regions. Additionally or alternatively, the high-transmittance can be achieved using touch electrode architecture techniques that use transparent or semi-transparent materials instead of opaque metal mesh within the high-transmittance regions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIGS. **1A-1E** illustrate example systems that can include a touch screen according to examples of the disclosure.

[0007] FIG. **2** illustrates an example computing system including a touch screen according to examples of the disclosure.

[0008] FIG. **3A** illustrates an example touch sensor circuit corresponding to a self-capacitance measurement of a touch node electrode and sensing circuit according to examples of the disclosure.

[0009] FIG. **3B** illustrates an example touch sensor circuit corresponding to a mutual-capacitance drive line and sense line and sensing circuit according to examples of the disclosure.

[0010] FIG. **4A** illustrates touch screen with touch electrodes arranged in rows and columns according to examples of the disclosure.

[0011] FIG. **4B** illustrates touch screen with touch node electrodes arranged in a pixelated touch node electrode configuration according to examples of the disclosure.

[0012] FIG. **5A** illustrates an example touch screen stack-up including a metal mesh layer according to examples of the disclosure.

[0013] FIG. **5B** illustrate a top view of a portion of a touch screen according to examples of the disclosure.

[0014] FIG. **6** illustrates an example unit cell that can be repeated across a touch sensor panel to form a bar-and-stripe pattern according to examples of the disclosure.

[0015] FIG. **7** illustrates an example of a touch sensor panel formed from unit cells according to examples of the disclosure.

[0016] FIG. **8** illustrates a metal mesh corresponding to a portion of unit cell according to examples of the disclosure.

[0017] FIGS. **9-12** illustrate portions of example touch screens including a high-transmittance region according to examples of the disclosure.

[0018] FIG. **13** illustrates a cross-sectional view of a portion of an example touch screen including a high-transmittance region according to examples of the disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0019] In the following description of examples, reference is made to the accompanying drawings which form a part hereof, and in which it is shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the disclosed examples.

[0020] This relates generally to touch sensor panels, and more particularly to touch electrode architectures for high-transmittance touch screens or touch screens including high-transmittance regions. In some examples, one or more optical devices can be integrated with a touch screen such that light associated with the one or more optical devices passes through one or more layers of the touch screen. In some such examples, to avoid degrading performance of the optical devices, one or more high-transmittance regions can be used. In some examples, the high-transmittance can be achieved using touch electrode architecture techniques that reduce or eliminate metal mesh within

the high-transmittance regions. Additionally or alternatively, the high-transmittance can be achieved using touch electrode architecture techniques that use transparent or semi-transparent materials instead of opaque metal mesh within the high-transmittance regions. As described herein, high-transmittance can refer to a transmittance above a threshold level (e.g., above 80% transmittance, above 85% transmittance, above 90% transmittance, above 95% transmittance, above 98% transmittance, etc.)

[0021] FIGS. **1A-1E** illustrate example systems that can include a touch screen according to examples of the disclosure. FIG. **1A** illustrates an example mobile telephone **136** that includes a touch screen **124** according to examples of the disclosure. FIG. **1B** illustrates an example digital media player **140** that includes a touch screen **126** according to examples of the disclosure. FIG. **1C** illustrates an example personal computer **144** that includes a touch screen **128** according to examples of the disclosure. FIG. **1D** illustrates an example tablet computing device **148** that includes a touch screen **130** according to examples of the disclosure. FIG. **1E** illustrates an example wearable device **150** that includes a touch screen **132** and can be attached to a user using a strap **152** according to examples of the disclosure. It is understood that a touch screen can be implemented in other devices as well.

[0022] In some examples, touch screens **124, 126, 128, 130** and **132** can be based on self-capacitance. A self-capacitance based touch system can include a matrix of small, individual plates of conductive material or groups of individual plates of conductive material forming larger conductive regions that can be referred to as touch electrodes or as touch node electrodes (as described below with reference to FIG. **4B**). For example, a touch screen can include a plurality of individual touch electrodes, each touch electrode identifying or representing a unique location (e.g., a touch node) on the touch screen at which touch or proximity is to be sensed, and each touch node electrode being electrically isolated from the other touch node electrodes in the touch screen/panel. Such a touch screen can be referred to as a pixelated self-capacitance touch screen, though it is understood that in some examples, the touch node electrodes on the touch screen can be used to perform scans other than self-capacitance scans on the touch screen (e.g., mutual capacitance scans). During operation, a touch node electrode can be stimulated with an alternating current (AC) waveform, and the self-capacitance to ground of the touch node electrode can be measured. As an object approaches the touch node electrode, the self-capacitance to ground of the touch node electrode can change (e.g., increase). This change in the self-capacitance of the touch node electrode can be detected and measured by the touch sensing system to determine the positions of multiple objects when they touch, or come in proximity to, the touch screen. In some examples, the touch node electrodes of a self-capacitance based touch system can be formed from rows and columns of conductive material, and changes in the self-capacitance to ground of the rows and columns can be detected, similar to above. In some examples, a touch screen can be multi-touch, single touch, projection scan, full-imaging multi-touch, capacitive touch, etc.

[0023] In some examples, touch screens **124, 126, 128, 130** and **132** can be based on mutual capacitance. A mutual capacitance based touch system can include electrodes arranged as drive and sense lines that may cross over each other on different layers (in a double-sided configuration), or may be adjacent to each other on the same layer (e.g., as described below with reference to FIG. **4A**). The crossing or adjacent locations can form touch nodes. During operation, the drive line can be stimulated with an AC waveform and the mutual capacitance of the touch node can be measured. As an object approaches the touch node, the mutual capacitance of the touch node can change (e.g., decrease). This change in the mutual capacitance of the touch node can be detected and measured by the touch sensing system to determine the positions of multiple objects when they touch, or come in proximity to, the touch screen. As described herein, in some examples, a mutual capacitance based touch system can form touch nodes from a matrix of small, individual plates of conductive material.

[0024] In some examples, touch screens **124, 126, 128, 130** and **132** can be based on mutual

capacitance and/or self-capacitance. The electrodes can be arranged as a matrix of small, individual plates of conductive material (e.g., as in touch node electrodes **408** in touch screen **402** in FIG. **4B**) or as drive lines and sense lines (e.g., as in row touch electrodes **404** and column touch electrodes **406** in touch screen **400** in FIG. **4A**), or in another pattern. The electrodes can be configurable for mutual capacitance or self-capacitance sensing or a combination of mutual and self-capacitance sensing. For example, in one mode of operation electrodes can be configured to sense mutual capacitance between electrodes and in a different mode of operation electrodes can be configured to sense self-capacitance of electrodes. In some examples, some of the electrodes can be configured to sense mutual capacitance therebetween and some of the electrodes can be configured to sense self-capacitance thereof.

[0025] FIG. **2** illustrates an example computing system including a touch screen according to examples of the disclosure. Computing system **200** can be included in, for example, a mobile phone, tablet, touchpad, portable or desktop computer, portable media player, wearable device or any mobile or non-mobile computing device that includes a touch screen or touch sensor panel. Computing system **200** can include a touch sensing system including one or more touch processors **202**, peripherals **204**, a touch controller **206**, and touch sensing circuitry (described in more detail below). Peripherals **204** can include, but are not limited to, random access memory (RAM) or other types of memory or storage, watchdog timers and the like. Touch controller **206** can include, but is not limited to, one or more sense channels **208**, channel scan logic **210** and driver logic **214**. Channel scan logic **210** can access RAM **212**, autonomously read data from the sense channels and provide control for the sense channels. In addition, channel scan logic **210** can control driver logic **214** to generate stimulation signals **216** at various frequencies and/or phases that can be selectively applied to drive regions of the touch sensing circuitry of touch screen **220**, as described in more detail below. In some examples, touch controller **206**, touch processor **202** and peripherals **204** can be integrated into a single application specific integrated circuit (ASIC), and in some examples can be integrated with touch screen **220** itself.

[0026] It should be apparent that the architecture shown in FIG. **2** is only one example architecture of computing system **200**, and that the system could have more or fewer components than shown, or a different configuration of components. In some examples, computing system **200** can include an energy storage device (e.g., a battery) to provide a power supply and/or communication circuitry to provide for wired or wireless communication (e.g., cellular, Bluetooth, Wi-Fi, etc.). The various components shown in FIG. **2** can be implemented in hardware, software, firmware or any combination thereof, including one or more signal processing and/or application specific integrated circuits.

[0027] Computing system **200** can include a host processor **228** for receiving outputs from touch processor **202** and performing actions based on the outputs. For example, host processor **228** can be connected to program storage **232** and a display controller/driver **234** (e.g., a Liquid-Crystal Display (LCD) driver). It is understood that although some examples of the disclosure may be described with reference to LCD displays, the scope of the disclosure is not so limited and can extend to other types of displays, such as Light-Emitting Diode (LED) displays, including Organic LED (OLED), Active-Matrix Organic LED (AMOLED) and Passive-Matrix Organic LED (PMOLED) displays. Display driver **234** can provide voltages on select (e.g., gate) lines to each pixel transistor and can provide data signals along data lines to these same transistors to control the pixel display image.

[0028] Host processor **228** can use display driver **234** to generate a display image on touch screen **220**, such as a display image of a user interface (UI), and can use touch processor **202** and touch controller **206** to detect a touch on or near touch screen **220**, such as a touch input to the displayed UI. The touch input can be used by computer programs stored in program storage **232** to perform actions that can include, but are not limited to, moving an object such as a cursor or pointer, scrolling or panning, adjusting control settings, opening a file or document, viewing a menu,

making a selection, executing instructions, operating a peripheral device connected to the host device, answering a telephone call, placing a telephone call, terminating a telephone call, changing the volume or audio settings, storing information related to telephone communications such as addresses, frequently dialed numbers, received calls, missed calls, logging onto a computer or a computer network, permitting authorized individuals access to restricted areas of the computer or computer network, loading a user profile associated with a user's preferred arrangement of the computer desktop, permitting access to web content, launching a particular program, encrypting or decoding a message, and/or the like. Host processor **228** can also perform additional functions that may not be related to touch processing.

[0029] Note that one or more of the functions described herein can be performed by firmware stored in memory (e.g., one of the peripherals **204** in FIG. **2**) and executed by touch processor **202**, or stored in program storage **232** and executed by host processor **228**. The firmware can also be stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. In the context of this document, a “non-transitory computer-readable storage medium” can be any medium (excluding signals) that can contain or store the program for use by or in connection with the instruction execution system, apparatus, or device. In some examples, RAM **212** or program storage **232** (or both) can be a non-transitory computer readable storage medium. One or both of RAM **212** and program storage **232** can have stored therein instructions, which when executed by touch processor **202** or host processor **228** or both, can cause the device including computing system **200** to perform one or more functions and methods of one or more examples of this disclosure. The computer-readable storage medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus or device, a portable computer diskette (magnetic), a random access memory (RAM) (magnetic), a read-only memory (ROM) (magnetic), an erasable programmable read-only memory (EPROM) (magnetic), a portable optical disc such a CD, CD-R, CD-RW, DVD, DVD-R, or DVD-RW, or flash memory such as compact flash cards, secured digital cards, USB memory devices, memory sticks, and the like.

[0030] The firmware can also be propagated within any transport medium for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. In the context of this document, a “transport medium” can be any medium that can communicate, propagate or transport the program for use by or in connection with the instruction execution system, apparatus, or device. The transport medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic or infrared wired or wireless propagation medium.

[0031] Touch screen **220** can be used to derive touch information at multiple discrete locations of the touch screen, referred to herein as touch nodes. Touch screen **220** can include touch sensing circuitry that can include a capacitive sensing medium having a plurality of drive lines **222** and a plurality of sense lines **223**. It should be noted that the term “lines” is sometimes used herein to mean simply conductive pathways, as one skilled in the art will readily understand, and is not limited to elements that are strictly linear, but includes pathways that change direction, and includes pathways of different size, shape, materials, etc. Drive lines **222** can be driven by stimulation signals **216** from driver logic **214** through a drive interface **224**, and resulting sense signals **217** generated in sense lines **223** can be transmitted through a sense interface **225** to sense channels **208** in touch controller **206**. In this way, drive lines and sense lines can be part of the touch sensing circuitry that can interact to form capacitive sensing nodes, which can be thought of as touch picture elements (touch pixels) and referred to herein as touch nodes, such as touch nodes

226 and **227**. This way of understanding can be particularly useful when touch screen **220** is viewed as capturing an “image” of touch (“touch image”). In other words, after touch controller **206** has determined whether a touch has been detected at each touch nodes in the touch screen, the pattern of touch nodes in the touch screen at which a touch occurred can be thought of as an “image” of touch (e.g., a pattern of fingers touching the touch screen). As used herein, an electrical component “coupled to” or “connected to” another electrical component encompasses a direct or indirect connection providing electrical path for communication or operation between the coupled components. Thus, for example, drive lines **222** may be directly connected to driver logic **214** or indirectly connected to driver logic **214** via drive interface **224** and sense lines **223** may be directly connected to sense channels **208** or indirectly connected to sense channels **208** via sense interface **225**. In either case an electrical path for driving and/or sensing the touch nodes can be provided. [0032] In some examples, computing system **200** can also include one or more optical devices **201**, which may also be referred to herein as optical components. In some examples, the one or more optical devices **201** can include components for light emitting and/or light sensing. In some examples, the one or more optical devices **201** can include light-emitting diodes (e.g., LEDs, OLEDs, etc.), cameras, lasers (e.g., vertical-cavity surface-emitting lasers, etc.), light detectors, photodiodes, and the like. In some examples, the operation of the optical devices can be controlled by host processor **228** or an optical controller (not shown) to perform functionality using the optical devices. The functionality can include, without limitation, projecting light, imaging, proximity sensing and ranging, ambient light sensing, photography, etc., among other possibilities. In some examples, the one or more optical devices **201** can be implemented in proximity to touch screen **220** (e.g., on a periphery of or a notch region along a perimeter of touch screen **220**). As described in more detail herein, in some examples, the one or more optical devices **201** can be integrated with touch screen **220** such that light passes through one or more layers of the touch screen. In some such examples, to avoid degrading performance of the optical devices **201**, a high-transmittance touch screen or a touch screen including one or more high-transmittance regions can be used. In some examples, the high-transmittance can be achieved using touch electrode architecture techniques described herein with respect to FIGS. **9-13**.

[0033] FIG. **3A** illustrates an example touch sensor circuit **300** corresponding to a self-capacitance measurement of a touch node electrode **302** and sensing circuit **314** according to examples of the disclosure. Touch node electrode **302** can correspond to a touch electrode **404** or **406** of touch screen **400** or a touch node electrode **408** of touch screen **402**. Touch node electrode **302** can have an inherent self-capacitance to ground associated with it, and also an additional self-capacitance to ground that is formed when an object, such as finger **305**, is in proximity to or touching the electrode. The total self-capacitance to ground of touch node electrode **302** can be illustrated as capacitance **304**. Touch node electrode **302** can be coupled to sensing circuit **314**. Sensing circuit **314** can include an operational amplifier **308**, feedback resistor **312** and feedback capacitor **310**, although other configurations can be employed. For example, feedback resistor **312** can be replaced by a switched capacitor resistor in order to minimize a parasitic capacitance effect that can be caused by a variable feedback resistor. Touch node electrode **302** can be coupled to the inverting input (−) of operational amplifier **308**. An AC voltage source **306** (V.sub.ac) can be coupled to the non-inverting input (+) of operational amplifier **308**. Touch sensor circuit **300** can be configured to sense changes (e.g., increases) in the total self-capacitance **304** of the touch node electrode **302** induced by a finger or object either touching or in proximity to the touch sensor panel. Output **320** can be used by a processor to determine the presence of a proximity or touch event, or the output can be inputted into a discrete logic network to determine the presence of a proximity or touch event.

[0034] FIG. **3B** illustrates an example touch sensor circuit **350** corresponding to a mutual-capacitance drive line **322** and sense line **326** and sensing circuit **314** according to examples of the disclosure. Drive line **322** can be stimulated by a stimulation signal output from AC voltage source

306 (e.g., an AC voltage signal). The stimulation signal can be capacitively coupled to sense line **326** through mutual capacitance **324** between drive line **322** and the sense line. When an object, such as finger **305**, approaches the touch node created by the intersection of drive line **322** and sense line **326**, mutual capacitance **324** can change (e.g., decrease). This change in mutual capacitance **324** can be detected to indicate a touch or proximity event at the touch node, as described herein. The sense signal coupled onto sense line **326** can be received by sensing circuit **314**. Sensing circuit **314** can include operational amplifier **308** and at least one of a feedback resistor **312** and a feedback capacitor **310**. FIG. 3B illustrates a general case in which both resistive and capacitive feedback elements are utilized. The sense signal (referred to as V_{in}) can be inputted into the inverting input of operational amplifier **308**, and the non-inverting input of the operational amplifier can be coupled to a reference voltage $V_{sub.ref}$. Operational amplifier **308** can drive its output to voltage $V_{sub.o}$ to keep $V_{sub.in}$ substantially equal to $V_{sub.ref}$, and can therefore maintain $V_{sub.in}$ constant or virtually grounded. A person of skill in the art would understand that in this context, equal can include deviations of up to 15%. Therefore, the gain of sensing circuit **314** can be mostly a function of the ratio of mutual capacitance **324** and the feedback impedance, comprised of resistor **312** and/or capacitor **310**. The output of sensing circuit **314** V_o can be filtered and heterodyned or homodyned by being fed into multiplier **328**, where V_o can be multiplied with local oscillator **330** to produce $V_{sub.detect}$. $V_{sub.detect}$ can be inputted into filter **332**. One skilled in the art will recognize that the placement of filter **332** can be varied; thus, the filter can be placed after multiplier **328**, as illustrated, or two filters can be employed: one before the multiplier and one after the multiplier. In some examples, there can be no filter at all. The direct current (DC) portion of $V_{sub.detect}$ can be used to determine if a touch or proximity event has occurred. Note that while FIGS. 3A-3B indicate the demodulation at multiplier **328** occurs in the analog domain, output V_o may be digitized by an analog-to-digital converter (ADC), and multiplier **328**, filter **332** and oscillator **330** may be implemented in a digital fashion (e.g., multiplier **328** can be a digital demodulator, filter **332** can be a digital filter, and oscillator **330** can be a digital NCO (Numerical Controlled Oscillator)).

[0035] Referring back to FIG. 2, in some examples, touch screen **220** can be an integrated touch screen in which touch sensing circuit elements of the touch sensing system can be integrated into the display pixel stack-ups of a display. The circuit elements in touch screen **220** can include, for example, elements that can exist in LCD or other displays (LED display, OLED display, etc.), such as one or more pixel transistors (e.g., thin film transistors (TFTs)), gate lines, data lines, pixel electrodes and common electrodes. In a given display pixel, a voltage between a pixel electrode and a common electrode can control a luminance of the display pixel. The voltage on the pixel electrode can be supplied by a data line through a pixel transistor, which can be controlled by a gate line. It is noted that circuit elements are not limited to whole circuit components, such as a whole capacitor, a whole transistor, etc., but can include portions of circuitry, such as only one of the two plates of a parallel plate capacitor. Additionally, as described herein, in some examples, the integrated touch screen can include optical devices and one or more high-transmittance regions corresponding to the optical devices.

[0036] FIG. 4A illustrates touch screen **400** with touch electrodes **404** and **406** arranged in rows and columns according to examples of the disclosure. Specifically, touch screen **400** can include a plurality of touch electrodes **404** disposed as rows, and a plurality of touch electrodes **406** disposed as columns. Touch electrodes **404** and touch electrodes **406** can be on the same or different material layers on touch screen **400**, and can intersect with each other, as illustrated in FIG. 4A. In some examples, the electrodes can be formed on opposite sides of a transparent (partially or fully) substrate and from a transparent (partially or fully) semiconductor material, such as ITO, though other materials are possible. Electrodes displayed on layers on different sides of the substrate can be referred to herein as a double-sided sensor. In some examples, the touch electrodes can be formed on the same layer, and may be referred to herein as a single-sided sensor. In some

examples, touch screen **400** can sense the self-capacitance of touch electrodes **404** and **406** to detect touch and/or proximity activity on touch screen **400**, and in some examples, touch screen **400** can sense the mutual capacitance between touch electrodes **404** and **406** to detect touch and/or proximity activity on touch screen **400**.

[0037] FIG. **4B** illustrates touch screen **402** with touch node electrodes **408** arranged in a pixelated touch node electrode configuration according to examples of the disclosure. Specifically, touch screen **402** can include a plurality of individual touch node electrodes **408**, each touch node electrode identifying or representing a unique location on the touch screen at which touch or proximity (i.e., a touch or proximity event) is to be sensed, and each touch node electrode being electrically isolated from the other touch node electrodes in the touch screen/panel, as previously described. Touch node electrodes **408** can be on the same or different material layers on touch screen **400**. In some examples, touch screen **402** can sense the self-capacitance of touch node electrodes **408** to detect touch and/or proximity activity on touch screen **402**, and in some examples, touch screen **402** can sense the mutual capacitance between touch node electrodes **408** to detect touch and/or proximity activity on touch screen **402**.

[0038] As described herein, in some examples, some or all of the touch electrodes of the touch screen can be formed from a metal mesh. FIG. **5A** illustrates an example touch screen stack-up including a metal mesh layer according to examples of the disclosure. Touch screen **500** can include a substrate **509** (e.g., a printed circuit board) upon which display LEDs **508** (optionally OLEDs) can be mounted. In some examples, the LEDs **508** can be partially or fully embedded in substrate **509** (e.g., the components can be placed in depressions in the substrate). Substrate **509** can include routing traces in one or more layers (e.g., represented by metal layer **510** in FIG. **5A**) to route the LEDs to display driving circuitry (e.g., display driver **234**). The stack-up of touch screen **500** can also include one or more passivation layers deposited over the LEDs **508**. For example, the stack-up of touch screen **500** illustrated in FIG. **5A** can include a passivation layer **507** (e.g., transparent epoxy) and passivation layer **517**. Passivation layers **507** and **517** can planarize the surface for respective metal mesh layers. Additionally, the passivation layers can provide electrical isolation (e.g., between metal mesh layers and between the LEDs and a metal mesh layer). Metal mesh layer **516** (e.g., copper, silver, etc.) can be deposited on the planarized surface of the passivation layer **517** over the display LEDs **508**, and metal mesh layer **506** (e.g., copper, silver, etc.) can be deposited on the planarized surface of passivation layer **507**. In some examples, the passivation layer **517** can include material to encapsulate the LEDs to protect them from corrosion or other environmental exposure. Metal mesh layer **506** and/or metal mesh layer **516** can include a pattern of conductor material in a mesh pattern described below. In some examples, metal mesh layer **506** and metal mesh layer **516** can be coupled by one or more vias. Additionally, although not shown in FIG. **5A**, a border region around the display active area can include metallization (or other conductive material) that may or may not be a metal mesh pattern. In some examples, metal mesh is formed of a non-transparent material, but the metal mesh wires are sufficiently thin and sparse to appear transparent to the human eye. The touch electrodes (and some routing) as described herein can be formed in the metal mesh layer(s) from portions of the metal mesh. In some examples, polarizer **504** can be disposed above the metal mesh layer **506** (optionally with another planarization layer disposed over the metal mesh layer **506**). Cover glass (or front crystal) **502** can be disposed over polarizer **504** and form the outer surface of touch screen **500**. It is understood that although two metal mesh layers (and two corresponding planarization layers) are illustrated, in some examples more or fewer metal mesh layers (and corresponding planarization layers) can be implemented. Additionally, it is understood that LEDs **508**, substrate **509**, metal layer **510**, and/or passivation layer **517** can be replaced by a thin-film transistor (TFT) LCD display (or other types of displays), in some examples. Additionally, it is understood that polarizer **504** can include one or more transparent layers including a polarizer, adhesive layers (e.g., optically clear adhesive) and protective layers.

[0039] FIG. 5B illustrates a top view of a portion of touch screen 500 in a diamond pattern according to examples of the disclosure. The top view shows metal mesh 540 (e.g., a portion of metal mesh layer 506) together with LEDs 508 of touch screen 500. The LEDs can be arranged in groups of three proximate LEDs, including a red LED (e.g., red LED 544), a green LED (e.g., green LED 546), and a blue LED (e.g., blue LED 548), to form standard red-green-blue (RGB) display pixels. Although primarily described herein in terms of an RGB display pixel, it is understood that other touch pixels are possible with different numbers of LEDs and/or different color LEDs. The metal mesh can be formed of conductors (e.g., metal mesh wires formed from conductive materials such as copper, silver, etc.) disposed in a pattern to allow light to pass (at least vertically) through the gaps in the mesh (e.g., the LEDs 508 can be disposed in the LED layer opposite openings in the metal mesh disposed in the metal mesh layer(s) 506 and/or 516). In other words, the conductors of metal mesh layer can be patterned so that conceptually flattening the metal mesh layer(s) and LEDs into the same layer, the conductors and the LEDs do not overlap. In some examples, the metal mesh wires in the metal mesh layer may overlap (at least partially) some of the LEDs 508, but may be thin enough or sparse enough to not obstruct a human's view of the LEDs. The metal mesh 540 can be formed in a diamond pattern around LEDs arranged in a diamond configuration. The pattern of LEDs forming the display pixels can be repeated across the touch screen to form the display. During fabrication, the metal mesh pattern can repeat across the touch screen to form a touch screen with uniform optical characteristics. It should be understood that the arrangement of LEDs and the corresponding metal mesh are merely an example, and other arrangements of LEDs and corresponding metal mesh patterns are possible. For example, the metal mesh can, in some examples, form a rectangular shape (or other suitable shape including polygonal shapes, etc.) around rectangular-shaped LEDs.

[0040] As described herein, the touch electrodes and/or routing can be formed from the metal mesh. To form the electrically isolated touch electrodes or electrically isolated groups of touch electrodes (e.g., groups of touch electrodes forming row electrodes or column electrodes), the metal mesh can be cut (e.g., chemically or laser etched, among other possibilities) to form a boundary between two adjacent touch electrodes, between two adjacent routing traces or between a routing trace and adjacent touch electrode. The cut in the metal mesh can electrically isolate the metal mesh forming a first touch electrode (or first group of touch electrodes) from the metal mesh forming a second touch electrode (or second group of touch electrodes). Similarly, cuts to the metal mesh can be made to electrically isolate the metal mesh forming a first touch electrode from a first routing trace or to electrically isolate the first routing trace from a second routing trace.

[0041] Referring back to FIG. 5A, in some examples, although metal mesh layers 506 and 516 are illustrated, it is understood that in some examples, alternative materials can be used in these layers to implement touch electrodes. Additionally, although LEDs 508 are described, it is understood that additional optical components apart from the display pixels (e.g., optical devices 201) can be implemented on substrate 509 (or another layer within the stack-up). Additionally, in some examples, a portion of one or more of metal mesh layers 506 and 516 may be used to implement touch electrodes or portions of one or more touch electrodes with a different material to improve the transmittance in regions corresponding to the optical components.

[0042] As described herein, in some examples, touch electrodes can be arranged in rows and columns formed in a first layer. In some examples, the touch electrodes can be arranged in a bar-and-stripe pattern. The column touch electrodes illustrated in FIGS. 6-7 can be referred to as “bars” and the row touch electrodes can be formed from interconnected touch electrode segments that can be referred to as “stripes” (e.g., interconnected via bridges). FIG. 6 illustrates an example unit cell that can be repeated across a touch sensor panel to form a bar-and-stripe pattern according to examples of the disclosure. FIG. 7 illustrates an example of a touch sensor panel formed of nine unit cells (3×3) corresponding to the example unit cell of FIG. 6 (with some modification). It is understood that other touch electrode patterns can be implemented within the scope of this

disclosure.

[0043] FIG. 6 illustrates an example unit cell corresponding to a touch node according to examples of the disclosure. The unit cell **600** can include a portion of a column touch electrode **602** (corresponding to a “bar”) and a portion of a row electrode formed from touch electrode segments **604A-604F** (corresponding to “stripes”). A mutual capacitance between the column touch electrode and the row touch electrode can change due to the proximity of an object (e.g., a finger) at a touch node corresponding to the unit cell. The column touch electrode **602** can correspond to a contiguous, electrically connected region, including regions around the touch electrode segments **604A-604F**. The touch electrode segments **604A-604F** of the row electrode can be electrically connected using one or more bridges **606A-606H** that bridge across the neck regions **608A-608D** of the column touch electrode **602** between the touch electrode segments **604A-604F**. In some examples, one bridge can be used to interconnect two touch electrode segments (e.g., bridges **606A-606D**). In some examples, more than one bridge can be used to interconnect two touch electrode segments (e.g., bridges **606A** and **606E**, bridges **606B** and **606F**, etc.). Bridge-connected touch electrodes segments **604A-604C** (e.g., corresponding to a first “stripe” in the bar-and-stripe pattern) and bridge-connected touch electrode segments **604D-604F** (e.g., corresponding to a second “stripe” in the bar-and-stripe pattern) can be electrically connected outside of the unit cell area (e.g., as illustrated in FIG. 7). In some examples, the first and second stripes can be electrically connected to one another within the unit cell area (e.g., with bridges). In some examples, bridges **606A-606H** may be achieved using wire bonds or other conductors formed without using a metal mesh layer (e.g., ITO, etc.). In some examples, bridges **606A-606H** may be formed using a metal mesh layer (e.g., metal mesh layer **516**) different than the metal mesh layer used to form column touch electrode **602** and touch electrode segments **604A-604F** (e.g., metal mesh layer **506**). The connection between the metal mesh layers can also include a via (or other interconnection), in some examples, to make connections between the first metal mesh layer and the second metal mesh layer. It is understood that that bridges **606A-606H** may include multiple metal mesh wires (e.g., increasing the width of the bridge) to meet the resistance requirements for the row touch electrodes.

[0044] The distribution of the touch electrode segments within the unit cell can improve the touch signal levels (and therefore the signal-to-noise ratio (SNR) for touch sensing) because mutual capacitance in a single-layer touch sensor panel can be a function of the distance between the touch electrodes that are driven and sensed. For example, the mutual capacitances can be greater along the boundaries between a touch electrode that is driven and a touch electrode that is sensed as compared with the center of the two touch electrodes. Thus, by dividing the row electrode into multiple stripes (thereby reducing the maximum spacing between a region of the drive electrode and a region of a sense electrode in the unit cell), the signal measured at the unit cell can be increased relative to other touch electrode patterns (e.g., a diamond touch electrode pattern, etc.). The impact of the distributed bar-and-stripe pattern on the mutual capacitance can provide increased modulation between finger and the sensor. Additionally, the distribution of the touch electrode segments can provide improved linearity of the touch signal detected as an object moves across the touch sensor panel (e.g., more uniform signal measured by an object, independent on the location of the object on the touch sensor panel). Improved linearity can provide various benefits of improved touch performance that include more precise and accurate touch location detection, reduced wobble, etc.

[0045] Although not shown in FIG. 6, in some examples, the unit cell can include buffering regions between portions of column touch electrode **602** and touch electrode segments of rows. The buffer regions can be conductive material that is floating (or grounded or driven with a potential, in some examples). The buffer region can reduce the baseline mutual capacitance of the touch node by increasing the distance between the drive and sense regions. For example, referring to unit cell **712** in FIG. 7, touch electrode segments forming rows in FIG. 6 can be separated on a first boundary with column touch electrode **702C** by buffer region **722A** and can be separated on a second

boundary with column electrode **702C** by buffer region **722B**. As illustrated in FIG. 7, similar buffer regions can be included between the column touch electrode and the touch electrode segments forming rows across the touch sensor panel. Although FIG. 7 illustrates buffer regions on two sides of each of the touch electrode segments, it is understood that in some examples, the buffering can be on fewer sides (one or no sides) or more sides (three or four sides) of the touch electrode segments. Increasing the separation (e.g., surface area and/or width) can further reduce the baseline capacitance, whereas decreasing the separation can increase the baseline capacitance. In some examples, as illustrated in FIG. 7, the neck region can be free of buffer regions to reduce the impedance of the column touch electrode. Additionally, although buffer regions are shown as continuous along a respective boundary of a touch electrode segment, it is understood that the buffer region can be discontinuous so as to be present in one or more segments along a portion of the boundary. Additionally, although similar buffer regions are shown on all touch electrode segments in unit cell **712**, it is understood that different touch electrode segments in a unit cell can have different numbers of buffer regions or buffer regions with different properties (dimensions, distributions, etc.).

[0046] Referring back to FIG. 6, it is understood that a unit cell can include fewer touch electrode segments and/or fewer interconnections between touch electrode segments. For example, four touch electrode segments can be connected using two interconnections, rather than connecting the six touch electrode segments of FIG. 6 with the four interconnections of FIG. 6. Reducing the number of interconnections can reduce the baseline mutual capacitance of the touch node because interconnections of bridges **606** can result in increased mutual capacitance due to the proximity between the drive and sense regions at these interconnections. Additionally, reducing the number of interconnections can reduce the resistance of the row touch electrodes. It should be understood that fewer or more interconnections and touch electrode segments can be employed.

[0047] It is understood that although FIG. 6 illustrates contiguous columns and segmented rows, in some examples, the column touch electrode can be formed from touch electrode segments that can be interconnected by bridges (e.g., in the neck region), and the row touch electrode can be formed from stripes, each of which can be contiguous (e.g., and may optionally be interconnected in the border area).

[0048] It should be understood that although unit cell **600** in FIG. 6 illustrates two stripes in the unit cell (two rows of interconnected touch electrode segments), that the number of stripes can be greater than two (e.g., three, four, etc.) or less than two (e.g., one) in some examples. It should be understood that unit cell **600** is an example unit cell. The number and dimensions of touch electrode segments, the number and dimensions of interconnections between touch electrode segments (and between portions of a column touch electrodes), and the thickness and dimensions of the neck region can be varied according to design considerations, including trading off the impedance of the row and/or column touch electrodes and the baseline capacitance for the unit cell, including an amount of desired touch signal, and including the linearity of the touch signals across the touch sensor panel. Although described separately above, one or more of the above described modifications of the unit cell can be combined in some examples. For example, the multiple bridges of FIG. 6 can be used with the buffer regions of FIG. 7. It should be understood that although column touch electrodes are illustrated as contiguous and row touch electrodes are illustrated as formed of touch electrode segments, in some examples, row touch electrodes can be contiguous and column touch electrodes can be formed of touch electrode segments.

[0049] FIG. 7 illustrates an example of a touch sensor panel formed from unit cells according to examples of the disclosure. For example, touch sensor panel **700** can include nine unit cells corresponding to unit cell **712** (3×3 touch nodes), which can be corresponding to the example unit cell of FIG. 6 (e.g., corresponding to unit cell **600**, modified to illustrate only a single bridge **706** between touch electrode segments **704** and to include buffer regions, such as buffer regions **722A** and **722B**). For brevity, the details of the unit cell are not repeated. As illustrated in FIG. 7, touch

sensor panel **700** can include three column touch electrodes **702A-702C** (“bars”) that can be driven during touch sensing operation (e.g., by drive signals provided by routing traces labeled “DRV_N-1”, “DRV_N”, and “DRV_N+1”). Touch sensor panel **700** can also include three row touch electrodes. Each of the row touch electrodes illustrated in FIG. 7 can include two “stripes” formed of touch electrode segments **704**. The touch electrode segments **704** for each “stripe” can be interconnected within the touch sensor panel active area (e.g., in the visible area of the display in a touch screen) by bridges **706** (e.g., metal mesh). Although one bridge **706** between touch electrode segments is illustrated in FIG. 7, it is understood that additional bridges can be used to improve electrostatic discharge protection, improve mechanical and/or electrical reliability of the connection and/or reduce impedance of the row touch electrode. Additionally, although not shown in FIG. 7, additional bridges can be used to provide the same or similar benefits for column touch electrodes. The two “stripes” of a row electrode can be connected in a border area (e.g., outside of the touch sensor panel active area/outside the visible area of the display) by conductive traces (e.g., metal mesh or otherwise). Each row electrode can be sensed during touch sensing operation (e.g., by sense channels coupled to routing traces labeled “SNS_N-1”, “SNS_N”, and “SNS_N+1”). The adjacencies of a respective column touch electrode and a respective row touch electrode can form a respective touch node/unit cell of touch sensor panel **700**.

[0050] Although the example unit cell **712** includes one or more buffer regions, it should be understood that alternative unit cells can be used, such as those described herein. For example, a unit cell may include only one “stripe”, or the multiple “stripes” of a unit cell may not be connected in a border area by conductive traces (e.g., one “stripe” per row electrode). Additionally, although a 3×3 grouping of unit cells is illustrated, it is understood that the panel can be of a smaller or larger size (e.g., 2×2, 4×4, 5×5, 10×10, 16×16, etc.) Additionally, although FIG. 7 illustrates column touch electrodes that are driven and rows touch electrodes that are sensed, in some examples, the row touch electrodes can be driven and the column touch electrodes can be sensed.

[0051] Although FIGS. 6-7 illustrate rectangular electrodes for row and column touch electrodes with linear boundaries, it should be understood that due to the pattern of metal mesh and to reduce the visibility of the metal mesh, the true shape of touch electrodes and their boundaries may not be rectangular. FIG. 8 illustrates a metal mesh corresponding to a portion of unit cell of FIG. 6 according to examples of the disclosure. Metal mesh **800** can correspond, for example, to half of unit cell **600** of FIG. 6. Metal mesh **800** can include a first metal mesh portion **802** corresponding to column touch electrode **602** and second metal mesh portions **804A-804C** corresponding to touch electrode segments **604A-604C**. Due to the diamond pattern (with 45 degree angles) and to reduce the visibility of the boundaries of the touch electrodes, the first and second metal mesh portions can be non-linear along the boundaries. In some examples, the boundaries between the touch electrodes can be a zig-zag or wave-like pattern. For example, as illustrated in FIG. 8, the boundary between first metal mesh portion **802** and second metal mesh portion **804B** can have a zig-zag pattern where the length of segments **812** and **814** can each be a length of three metal mesh wires. A similar pattern can be implemented for the other boundaries illustrated in FIG. 8 (with slight variations at the corners for continuity according to the geometry of the pattern). It should be understood that the length of segments **812** and **814** are exemplary, and other lengths are possible. Additionally, the lengths can be different at different points along a boundary or different between two different boundaries. In some examples, rather than defining the pattern by the lengths of segments such as segments **812** and **814**, the zig-zag pattern can be defined by other parameters.

[0052] The touch electrodes (and buffer regions) can be formed from metal mesh in the metal mesh layer (e.g., corresponding to metal mesh layer **506**) by cuts or electrical discontinuities in the metal mesh wires between the touch electrodes (and/or buffer regions). In some examples, the cuts or electrical discontinuities can be formed at midpoints of metal mesh wires (or otherwise dividing one or more metal mesh wires), rather than having cuts or electrical discontinuities at vertices of two metal mesh wires in the metal mesh pattern.

[0053] In some examples, dummy cuts can further reduce visibility of the metal mesh boundary cuts. A dummy cut can interrupt one electrical path between two portions of the metal mesh (on either side of the dummy cut), without electrically isolating the metal mesh due to one or more other electrical paths between two portions of the metal mesh (on either side of the dummy cut). In other words, the portions of the metal mesh can remain at substantially the same electrical potential despite the internal cuts because the portions of the metal mesh are electrically connected. For example, dummy cuts can be made within the first metal mesh portion **802** and/or in the second metal mesh portions **804A-804C** that form physical separations in the metal mesh without electrically separating the metal mesh in each respective portion. In some examples, the dummy cuts can form a pattern that can be repeated across each of the touch electrodes. For example, a dummy cut unit (e.g., a pattern of discontinuities) can be defined, and the dummy cut unit can be repeated across the touch screen to form the dummy cuts. In some examples, dummy cuts can also be implemented for buffer regions (e.g., buffer region **722A-722B**) between the column touch electrodes and touch electrode segments.

[0054] In some examples, dummy cuts in the first metal mesh portion **802** can be restricted to certain regions. For example, dummy cuts may be excluded, or limited, in neck regions **808** of the first metal mesh portion **802**. Excluding (or limiting) dummy cuts in the neck regions **808** can be beneficial in some instances to reduce the impedance of the column touch sensor (due to the narrow width of the metal mesh in the neck regions).

[0055] Although FIGS. **6-8** illustrate column touch electrodes and row touch electrodes disposed in a first metal mesh layer (e.g., corresponding to metal mesh layer **506**) that may include interconnections in a second metal mesh layer (e.g., corresponding to metal mesh layer **516**), it should be understood that in some examples, the column touch electrodes can be disposed in one layer and the row touch electrodes can be disposed in another layer (e.g., in a double-sided touch sensor configuration as illustrated in FIG. **4A**).

[0056] As described herein, in some examples, touch electrode architectures can be improved for high-transmittance touch screens or touch screens including high-transmittance regions. For example, one or more optical devices (e.g., optical device(s) **201**) can be integrated with touch screen **220** such that light passes through one or more layers of the touch screen. Improved transmittance can improve performance of the optical devices.

[0057] FIGS. **9-12** illustrates portions of example touch screens including a high-transmittance region according to examples of the disclosure. Although FIGS. **9-12** show a circular high-transmittance region, it is understood that the high-transmittance region may have a different shape (e.g., square, rectangular, a shape corresponding to the geometry of the optical components, a non-regular geometric shape, etc.). For example, region **900** in FIG. **9** can correspond to a view of a region **720** of touch sensor panel **700** shown in FIG. **7** including two column touch electrodes and two row touch electrodes (or two stripes of a single row electrode). Region **900** of a touch sensor panel can include portions of a first column touch electrode **902A** including neck regions **908A** and **908C** and a second column touch electrode **902B** including neck regions **908B** and **908D** (e.g., corresponding to column touch electrodes **602A-602B**, **702A-702B**, and neck regions **608A-608D**), portions of touch electrode segments **904A-904C** interconnected using bridges **906A-906D** (e.g., corresponding touch electrode segments **604A-604C**, **704** interconnected using bridges **606A**, **606B**, **606E**, **606F**, **706**), and portions of touch electrode segments **904D-904F** interconnected using bridges **906E-906H** (e.g., corresponding touch electrode segments **604D-604F**, **704** interconnected using bridges **606C**, **606D**, **606G**, **606H**, **706**). As described herein, the touch electrodes and bridges can be formed of metal mesh in one or more metal mesh layer s (e.g., touch electrodes in a first metal mesh layer and bridges in a second metal mesh layer). Region **910** (e.g., corresponding to region **710**) indicates a region that requires improved transmittance of the touch electrodes due to one or more optical components. Region **910** can intersect the first row electrode (e.g., corresponding to touch electrode segments **904A-904C**) and not intersect the second row electrode

(e.g., corresponding to touch electrode segments **904D-904F**). As illustrated in FIG. 9, touch electrode segments **904A-904C** (also referred to herein as a first touch electrode segment, a second touch electrode segment, and a third touch electrode segment) are aligned on the horizontal axis with touch electrode segments **904D-904F** (also referred to herein as a fourth touch electrode segment, a fifth touch electrode segment, and a sixth touch electrode segment). Additionally, as illustrated in FIG. 9, neck regions **908A** and **908C** of column touch electrode **902A** are aligned on the horizontal axis, and neck regions **908B** and **908D** of column touch electrode **902B** are aligned on the horizontal axis.

[0058] In some examples, to improve transmittance of the touch screen in a region corresponding to optical components (e.g., region **910**), bridges can be removed from the region. FIG. 10 illustrates a region **1000** of a touch sensor panel (e.g., corresponding to region **900**) that can include portions of a first column touch electrode **1002A** including neck regions **1008A** and **1008C** and a second column touch electrode **1002B** including neck regions **1008B** and **1008D** (e.g., corresponding to column touch electrodes **902A-902B** and neck regions **908A-908D**), portions of touch electrode segments **1004A-1004C** interconnected using bridges **1006A-1006D** (e.g., corresponding touch electrode segments **904A-904C** interconnected using bridges **906A-906D**), and portions of touch electrode segments **1004D-1004F** interconnected using bridges **1006E-1006H** (e.g., corresponding touch electrode segments **904D-904F** interconnected using bridges **906E-906H**). Region **1010** (e.g., corresponding to region **910**) indicates a region that requires improved transmittance of the touch electrodes due to one or more optical components. Region **1010** can intersect the first row electrode (e.g., corresponding to touch electrode segments **1004A-1004C**) and not intersect the second row electrode (e.g., corresponding to touch electrode segments **1004D-1004F**).

[0059] As shown in FIG. 10, bridges **1006A-1006D** are outside region **1010**, unlike bridges **906A-906D** of FIG. 9, which are partially or fully within region **910**. Moving the bridges outside of region **1010** can be achieved by modifying the touch electrode pattern for the area at and around region **1010**. Specifically, touch electrode segments **1004A** and **1004C** (e.g., corresponding to a first row electrode) can have increased separation compared with touch electrode segments **1004D** and **1004F** (e.g., corresponding to a second row electrode that does not intersect a high-transmittance region) (e.g., d_2 of FIG. 10 is greater than d_1 of FIG. 9), and touch electrode segment **1004B** can have increased width compared with touch electrode segment **904B** (e.g., W_2 of FIG. 10 is greater than W_1 of FIG. 9). Another way of viewing the modification of FIG. 10 is as a shifting of the neck region **1008A** and **1008B** for the column touch electrodes/bars outside region **1010** compared with the neck regions **908A** and **908B** shown partially or fully within region **910** in FIG. 9. As a result, the neck regions **1008A** and **1008C** for column touch electrode **1002A** are offset on the horizontal axis, and the neck regions **1008B** and **1008D** for column touch electrode **1002B** are offset on the horizontal axis. As a result of the modification, touch electrode segments **1004A-1004C** are only partially aligned with touch electrode segments **1004D-1004F** along the horizontal axis.

[0060] Shifting the bridges outside of the high-transmittance region can allow for the remaining metal mesh of the touch electrodes in region **1010** to have uniformity within the region for improved optical performance (whereas the bridges create some non-uniformities in a different metal mesh layer). Additionally, shifting the bridges out of the high-transmittance region can allow for maintaining the resistance of the column touch electrode (e.g., there is no change in the dimensions of the neck regions **1008A-1008D** for the column touch electrodes compared with the dimensions of the neck regions **908A-908D**).

[0061] It is understood that in some examples, the bridges may be only partially moved out of region **1010**, and that partial movement of the bridges can also provide partial improvement in the transmittance. It is also understood that, in some examples, the changes to touch electrode segments **1004A-1004C** in FIG. 10 can be applied to other regions of the touch sensor panel (e.g.,

to touch electrode segments **1004D-1004F**) for pattern uniformity.

[0062] In some examples, in addition to removing the bridges from a region of the panel corresponding to an optical component, metal mesh of the touch electrodes can be removed from the region. FIG. **11** illustrates region **1100** of a touch sensor panel (e.g., corresponding to regions **900**, **1000**) that includes portions of a first column touch electrode **1102A** and a second column touch electrode **1102B** (e.g., corresponding to column touch electrodes **902A-902B**, **1002A-1002B**). FIG. **11** also illustrates portions of touch electrode segments **1104A-1104C** interconnected using bridges **1106A-1106D** (e.g., corresponding touch electrode segments **904A-904C**, **1004A-1004C** interconnected using bridges **906A-906D**, **1006A-1006D**), and portions of touch electrode segments **1104D-1104F** interconnected using bridges **1106E-1106H** (e.g., corresponding touch electrode segments **904D-904F**, **1004A-1004F** interconnected using bridges **906E-906H**, **1006E-1006H**). Region **1110** (e.g., corresponding to region **910**, **1010**) indicates a region that requires improved transmittance of the touch electrodes due to one or more optical components, which can intersect the first row electrode (e.g., corresponding to touch electrode segments **1104A-1104C**) and not intersect the second row electrode (e.g., corresponding to touch electrode segments **1104D-1104F**).

[0063] As shown in FIG. **11**, bridges **1106A-1106D** are outside region **1110**, unlike bridges **906A-906D** of FIG. **9**, which are partially or fully within region **910**. Additionally, touch electrode segment **1104B** (e.g., on the perimeter of region **1110**) and column touch electrodes **1102A-1102B** can be disposed outside of region **1110** (e.g., on the perimeter of touch electrode segment **1104B**). Moving the bridges and touch electrodes outside of region **1110** can be achieved by modifying the touch electrode pattern for the area at and around region **1110**. Specifically, touch electrode segments **1104A** and **1104C** (e.g., corresponding to a first row electrode) can have increased separation compared with touch electrode segments **1104D** and **1104F** (e.g., corresponding to a second row electrode that does not intersect a high-transmittance region), and in some examples, compared with touch electrode segments **1004A** and **1004C** (e.g., d3 of FIG. **11** is greater than d1 of FIG. **9**, and possibly d3 of FIG. **11** is greater than d2 of FIG. **10**). The pattern of column touch electrodes **1102A-1102B** and touch electrode segment **1104B** can be modified to remove any patterning from region **1110**. For example, as shown in FIG. **11**, touch electrode segment **1104B** optionally circumscribes region **1110** and column touch electrodes **1102A-1102B** optionally together circumscribe touch electrode segment **1104B**. In some examples, touch electrode segments **1104B** can be designed to have dimensions that allow for the first row electrode including touch electrode segments **1104B** to have the same resistance as (or within a threshold of, such as 1%, 5%, 10%, etc.) the second row electrode including touch electrode segment **1104E**. In some examples, touch electrode segments **1104B** can have an area that is the same as (or within a threshold of, such as 1%, 5%, 10%, etc.) the area of touch electrode segment **1104E**, such that the capacitive coupling for the touch electrode segments **1104B** and **1104E** can be the same (or within a threshold). In some examples, touch electrode segments **1104B** can be disc-shaped, where an inner diameter of the disc matches the border of region **1110**. It is understood that the shape of touch electrode segment **1104B** (e.g., the inner dimensions) may be different when region **1110** has a different shape than shown. Additionally, it is understood that the inner dimensions and outer dimensions of touch electrode segment can be different (e.g., circular inner dimensions, polygonal outer dimensions, etc.) Additionally, it is understood that touch electrode segment **1104B** may not fully circumscribe region **1110**. For example, an arc or half-circle can be used to implement touch electrode segment **1104B** to create a path between touch electrode segment **1104A** and **1104B** (using two or more of bridges **1106A-1106D**).

[0064] Shifting the bridges and touch electrodes outside of the high-transmittance region can allow for improved optical performance by removing the non-transparent or opaque metal mesh. Additionally, the touch electrode architecture allows for maintaining connections to form the column touch electrodes (e.g., using the neck regions) and row touch electrodes (e.g., using

bridges) outside region **1110**. As shown in FIG. **11**, the connections to form the row electrode can be achieved by routing the touch electrode segment **1104B** around region **1110** in one metal mesh layer (e.g., metal mesh layer **506**) along with the other touch electrodes, and using bridges in another metal mesh layer (e.g., metal mesh layer **516**) to bridge the neck region of the column touch electrodes.

[0065] It is understood that, in some examples, the bridges and/or touch electrodes may be only partially moved out of region **1110**, and that partial movement of the bridges and/or touch electrodes can also provide partial improvement in the transmittance. It is also understood that, in some examples, the changes to touch electrode segments **1104A-1104C** in FIG. **11** can be applied to other regions of the touch sensor panel (e.g., for pattern uniformity).

[0066] FIG. **12** illustrates region **1200** of a touch sensor panel (e.g., corresponding to regions **900**, **1000**, **1100**) that includes portions of a first column touch electrode **1202A** and a second column touch electrode **1202B** (e.g., corresponding to column touch electrodes **902A-902B**, **1002A-1002B**, **1102A-1102B**). FIG. **12** also illustrates portions of touch electrode segments **1204A** and **1204C** interconnected using bridges **1206A-1206B**, and portions of touch electrode segments **1204D-1204F** interconnected using bridges **1206E-1206H** (e.g., corresponding touch electrode segments **904D-904F**, **1004D-1004F**, **1104D-1104F** interconnected using bridges **906E-906H**, **1006E-1006H**, **1106E-1106H**). Region **1210** (e.g., corresponding to region **910**, **1010**, **1110**) indicates a region that requires improved transmittance of the touch electrodes due to one or more optical components, which can intersect the first row electrode (e.g., corresponding to touch electrode segments **1204A** and **1204C**) and not intersect the second row electrode (e.g., corresponding to touch electrode segments **1204D-1204F**).

[0067] As shown in FIG. **12**, bridges **1206A-1206B** are outside region **1210**, unlike bridges **906A-906D** of FIG. **9**, which are partially or fully within region **910**. Additionally, column touch electrodes **1202A-1202B** can be disposed outside of region **1210** (e.g., on the perimeter of region **1210**). A touch electrode segment that would otherwise have been located between touch electrode segments **1204A** and **1204C** (e.g., like touch electrode segments **904B**, **1004B**, **1104B**) is removed entirely from region **1200**. Instead, bridges **1206A** and **1206B** are implemented as longer bridges compared with bridges **906A-906D**, **1006A-1006D**, **1106A-1106D**. Bridges can be routed around the outside of region **1210** in a different layer than the touch electrodes. In some examples, region **1200** can represent an appearance of a portion of a touch sensor panel corresponding to a first row electrode and region **900** can represent an appearance of a different portion of the touch sensor panel corresponding to a second row electrode, but without including high-transmittance region **910** (e.g., with regions **900** and **1200** being horizontally aligned in the touch sensor panel). Like the touch electrode architecture of FIG. **11**, the touch electrode architecture of FIG. **12** can also have increased separation between touch electrode segments **1204A** and **1204C** (e.g., corresponding to a first row electrode) compared with touch electrode segments **1204D** and **1204F** (e.g., corresponding to a second row electrode that does not intersect a high-transmittance region), and, in some examples, compared with touch electrode segments **1004A** and **1004C** (e.g., d_3 of FIG. **12** is greater than d_1 of FIG. **9**, and possibly d_3 of FIG. **12** is greater than d_2 of FIG. **10**). It is understood that the shape of bridges **1206A-1206B** may be different when region **1210** has a different shape than shown. Additionally, it is understood that more and/or thicker bridges can be used in the touch electrode architecture of FIG. **12** compared with the bridges of the touch electrode architectures of FIGS. **9-11** to compensate for the length of bridges **1206A-1206B** compared with the bridges of FIGS. **9-11**. It is also understood that bridges may be implemented in a manner in which the bridges do not fully circumscribe region **1210**. For example, a half-circle can be used to implement bridge **1206A** to create a path between touch electrode segment **1204A** and **1204C** along half the circumference of region **1210** without using a bridge along the second half of the circumference of region **1210**.

[0068] Shifting the bridges and touch electrodes outside of the high-transmittance region can allow

for improved optical performance by removing the non-transparent or opaque metal mesh. Additionally, the touch electrode architecture allows for maintaining connections to form the column touch electrodes (e.g., using the neck regions) and row touch electrodes (e.g., using bridges) outside region **1210**. As shown in FIG. **12**, the connections to form the row electrode can be achieved by routing the bridges **1206A-1206B** around region **1210** in one metal mesh layer (e.g., metal mesh layer **516**) different than another metal mesh layer (e.g., metal mesh layer **506**) in which the other touch electrodes are implemented.

[0069] It is understood that in some examples, the bridges may be only partially moved out of region **1210**, and that partial movement of the bridges can also provide partial improvement in the transmittance. It is also understood that, in some examples, the changes to touch electrode segments **1204A** and **1204C** and bridges **1206A-1206B** in FIG. **12** can be applied to other regions of the touch sensor panel (e.g., for pattern uniformity). In some examples, bridges **1206A** and **1206B** can be used in touch electrode architecture of FIG. **11** in place of bridges **1106A-1106D** (e.g., to reduce the line resistance of the row electrode).

[0070] In some examples, transmittance can be improved by using transparent or partially transparent material(s) in place of opaque materials to form touch electrodes and/or bridges in a region that requires high-transmittance. Using a transparent or partially transparent material(s) in place of opaque materials can allow for touch electrode uniformity across the touch sensor panel for improved touch performance (e.g., better uniformity of touch signal across the panel) as well as improving the transmittance required for the region due to optical components. Referring back to FIG. **9** (or similarly FIG. **10**), some or all of the portions of the first column touch electrode **902A**, the second column touch electrode **902B**, the touch electrode segments **904B-904C**, and/or bridges **906A-906D** that overlap with region **910** can be implemented using transparent or semi-transparent materials, whereas outside of regions **910** (and outside multiple similar regions when multiple high-transmittance regions are implemented) the touch electrodes and/or bridges can be implemented with opaque materials (e.g., metal mesh). In some examples, the portions of the touch electrodes (e.g., first column touch electrode **902A**, second column touch electrode **902B**, touch electrode segments **904B-904C**) can be implemented using transparent or semi-transparent materials, whereas bridges **906A-906D** can be implemented using opaque materials (e.g., optionally by shifting bridges outside of high-transmittance region as in the architecture of FIG. **10**).

[0071] In some examples, the transparent or semi-transparent material can include Indium Tin Oxide (ITO). In some examples, the transparent or semi-transparent material can include conductive polymers, graphene, nanowires (e.g., silver nanowires), or nanotubes (e.g., carbon nanotubes). In some examples, the transparent or semi-transparent material(s) used to implement the portions of the touch electrodes can be implemented in the same layer in the stack-up as opaque metal mesh. For example, metal mesh layer **506** in the stack of touch screen **500** can represent a layer in which opaque metal mesh and a transparent or semi-transparent material can be disposed, with the transparent or semi-transparent material(s) disposed in the high-transmittance region(s) and with the opaque metal mesh be disposed outside the high-transmittance region(s). In some examples, within the high-transmittance region(s) the transparent or semi-transparent material(s) can be patterned in a similar manner (e.g., mesh) as the opaque metal mesh material forming touch electrodes outside of the high-transmittance region(s). In some examples, within the high-transmittance region(s) the transparent or semi-transparent material(s) can be patterned in a different manner as the opaque metal mesh material forming touch electrodes outside of the high-transmittance region(s). In some examples, within the high-transmittance region(s) the transparent or semi-transparent material(s) can be solid (e.g., not a mesh pattern), unlike the mesh pattern of touch electrodes outside of the high-transmittance region(s).

[0072] In some examples, the transparent or semi-transparent material(s) forming touch electrodes and/or bridges can be formed in a different layer than the touch electrodes formed from opaque

metal mesh (e.g., a layer above or below metal mesh layer **506**). FIG. **13** illustrates a cross-sectional view of a portion of an example touch screen including a high-transmittance region according to examples of the disclosure. The cross-sectional view **1300** of FIG. **13** is along the line AA' shown in FIG. **9** through region **910**. FIG. **13** illustrates portions of column electrodes **1302A** and **1302B** in a first layer outside of high-transmittance region **1308** (e.g., corresponding to portions of column electrodes **902A-902B** and region **910** in FIG. **9**). FIG. **13** also illustrates portions of column touch electrodes **1302C** and **1302D** in a different region within high-transmittance region **1308** (e.g., corresponding to other portions of column electrodes **902A-902B**). In some examples, portions of column touch electrodes **1302C** and **1302D** partially extend outside of high-transmittance region **1308** to enable coupling between the portions of column electrodes in the two layers outside of region **1308**. For example, as shown in FIG. **13**, a via **1310A** (or multiple vias) can connect touch electrode **1302A** and **1302C** outside region **1308**, and a via **1310B** (or multiple vias) can connect touch electrode **1302B** and **1302D** outside region **1308**. In some examples, connections between the touch electrodes in the two layers can be achieved using via-like connections within region **1308** (e.g., near the perimeter of region **1308**). In some examples, vias **1310A-1310B** can be omitted so that touch electrodes **1302C** and **1302D** electrically float. In such examples, touch electrodes **1302A** and **1302C** can be coupled capacitively and touch electrodes **1302B** and **1302D** can be coupled capacitively for use in touch sensing.

[0073] Although primarily described in the context of high-transmittance region(s), it is understood that transparent material may be used for touch electrodes across the touch sensor panel (e.g., both within and outside high-transmittance regions corresponding to optical components).

[0074] It is understood that although the touch electrode architectures of FIGS. **9-13** are described separately, that features can be combined in some examples. For example, the use of transparent material in the high-transmittance region(s) can be applied to the touch electrode architecture of FIG. **9** or **10**. Additionally or alternatively, the routing of the touch electrode architecture of FIG. **12** can be used in combination with the touch electrode architecture (including touch electrode segment **1104B**) of FIG. **11**. It is understood that although the touch electrode architectures of FIGS. **9-13** use the bar-and-stripe design of unit cells **600** or **712**, that other patterning can be used, and that the boundaries of the touch electrodes and/or routing may not be rectangular (e.g., as described with reference to FIG. **8**). It is also understood that although FIGS. **9-13** show the high-transmittance region overlapping two columns and one row, that the high transmittance region can overlap fewer or more columns and/or more rows.

[0075] Therefore, according to the above, some examples of the disclosure are directed to a touch screen. The touch screen can comprise: a display having an active area; an optical device in the active area at a position corresponding to a first region; and a plurality of touch electrodes formed of metal mesh disposed in a first metal mesh layer disposed over the active area of the display. The plurality of touch electrodes can include: a plurality of contiguous column touch electrodes including a first column touch electrode and a second column touch electrode; and a plurality of row touch electrodes including a first row touch electrode and a second row touch electrode. The first row touch electrode can be formed from a first plurality of touch electrode segments interconnected by a first plurality of bridges formed at least partially in a second metal mesh layer different from the first metal mesh layer. The second row touch electrode can be formed from a second plurality of touch electrode segments interconnected by a second plurality of bridges formed at least partially in the second metal mesh layer. The first plurality of touch electrode segments includes a first touch electrode segment and a second touch electrode segment. The first region is between the first touch electrode segment and the second touch electrode segment. The second plurality of touch electrode segments includes a fourth touch electrode segment, a fifth touch electrode segment, and a sixth touch electrode segment. The fourth touch electrode segment, the fifth touch electrode segment, and the sixth touch electrode segment can be consecutive within the second row touch electrode. A distance between the first touch electrode segment and the

second touch electrode segment can be greater than a distance between the fourth touch electrode segment and the sixth touch electrode segment.

[0076] Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first plurality of touch electrode segments can include a third touch electrode segment that is outside the first region. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the third touch electrode segment can correspond to the first column touch electrode and the second column touch electrode. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first plurality of bridges can include a first bridge between the first touch electrode segment and the third touch electrode segment and a second bridge between the second touch electrode segment and the third touch electrode segment. The first bridge and the second bridge can be outside the first region.

Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first bridge and the second bridge together circumscribe the first region. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the third touch electrode segment circumscribes the first region.

[0077] Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first plurality of bridges can include a first bridge between the first touch electrode segment and the second touch electrode segment and a second bridge between the first touch electrode segment and the second touch electrode segment. The first bridge and the second bridge can be outside the first region. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first bridge and the second bridge together circumscribe the first region.

[0078] Additionally or alternatively to one or more of the examples disclosed above, in some examples, a first neck region of the first column touch electrode corresponding to the first row touch electrode is offset on a horizontal axis from a second neck region of the first column touch electrode corresponding to the second row touch electrode.

[0079] Some examples of the disclosure are directed to a touch-sensitive device. The touch-sensitive device can comprise: an energy storage device; communication circuitry; a touch controller; and a touch screen. The touch screen can comprise: a display having an active area; an optical device in the active area at a position corresponding to a first region; and a plurality of touch electrodes formed of metal mesh disposed in a first metal mesh layer disposed over the active area of the display. The plurality of touch electrodes can include: a plurality of contiguous column touch electrodes including a first column touch electrode and a second column touch electrode; and a plurality of row touch electrodes including a first row touch electrode and a second row touch electrode. The first row touch electrode can be formed from a first plurality of touch electrode segments interconnected by a first plurality of bridges formed at least partially in a second metal mesh layer different from the first metal mesh layer. The second row touch electrode can be formed from a second plurality of touch electrode segments interconnected by a second plurality of bridges formed at least partially in the second metal mesh layer. The first plurality of touch electrode segments includes a first touch electrode segment and a second touch electrode segment. The first region is between the first touch electrode segment and the second touch electrode segment. The second plurality of touch electrode segments includes a fourth touch electrode segment, a fifth touch electrode segment, and a sixth touch electrode segment. The fourth touch electrode segment, the fifth touch electrode segment, and the sixth touch electrode segment can be consecutive within the second row touch electrode. A distance between the first touch electrode segment and the second touch electrode segment can be greater than a distance between the fourth touch electrode segment and the sixth touch electrode segment.

[0080] Some examples of the disclosure are directed to a touch screen. The touch screen can comprise a display having an active area; an optical device in the active area at a position corresponding to a first region; and a plurality of touch electrodes disposed in a first layer disposed

over the active area of the display. The plurality of touch electrodes can include: a plurality of column touch electrodes including a first column touch electrode and a second column touch electrode, the first column touch electrode and the second column touch electrode formed at least partially of metal mesh; and a plurality of row touch electrodes formed at least partially of metal mesh. A first row touch electrode of the plurality of row touch electrodes can be formed from a plurality of touch electrode segments interconnected by a first plurality of bridges formed at least partially in a second layer different from the first layer. A first portion of the first column touch electrode, a first portion of the second column touch electrode, or a first portion of the at least one of the plurality of touch electrode segments within the first region can be formed from a transparent material. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first portion of the first column touch electrode, the first portion of the second column touch electrode, or the first portion of the at least one of the plurality of touch electrode segments within the first region that is formed from the transparent material can be patterned with a first pattern. Additionally or alternatively to one or more of the examples disclosed above, in some examples, a second portion of the first column touch electrode, a second portion of the second column touch electrode, or a second portion of the at least one of the plurality of touch electrode segments outside the first region can be formed from the metal mesh patterned with the first pattern. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first portion of the first column touch electrode, the first portion of the second column touch electrode, or the first portion of the at least one of the plurality of touch electrode segments within the first region that is formed from the transparent material can be solid. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first portion of the first column touch electrode, the first portion of the second column touch electrode, or the first portion of the at least one of the plurality of touch electrode segments within the first region can be disposed in the second layer or a third layer different from the first layer and the second layer. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first portion of the first column touch electrode disposed in the second layer or the third layer can be coupled to a second portion of the first column touch electrode outside the first region using a first via, the first portion of the second column touch electrode disposed in the second layer or the third layer can be coupled to a second portion of the second column touch electrode outside the first region using a second via, or the first portion of the at least one of the plurality of touch electrode segments disposed in the second layer or the third layer can be coupled to a second portion of the at least one of the plurality of touch electrode segments outside the first region using a third via. Additionally or alternatively to one or more of the examples disclosed above, in some examples, the first portion of the first column touch electrode disposed in the second layer or the third layer can be capacitively coupled to a second portion of the first column touch electrode outside the first region, the first portion of the second column touch electrode disposed in the second layer or the third layer can be capacitively coupled to a second portion of the second column touch electrode outside the first region, or the first portion of the at least one of the plurality of touch electrode segments disposed in the second layer or the third layer can be capacitively coupled to a second portion of the at least one of the plurality of touch electrode segments outside the first region.

[0081] Some examples of the disclosure are directed to a touch-sensitive device. The touch-sensitive device can comprise: an energy storage device; communication circuitry; a touch controller; and a touch screen. The touch screen can comprise a display having an active area; an optical device in the active area at a position corresponding to a first region; and a plurality of touch electrodes disposed in a first layer disposed over the active area of the display. The plurality of touch electrodes can include: a plurality of column touch electrodes including a first column touch electrode and a second column touch electrode, the first column touch electrode and the second column touch electrode formed at least partially of metal mesh; and a plurality of row touch

electrodes formed at least partially of metal mesh. A first row touch electrode of the plurality of row touch electrodes can be formed from a plurality of touch electrode segments interconnected by a first plurality of bridges formed at least partially in a second layer different from the first layer. A first portion of the first column touch electrode, a first portion of the second column touch electrode, or a first portion of the at least one of the plurality of touch electrode segments within the first region can be formed from a transparent material.

[0082] Although examples of this disclosure have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of examples of this disclosure as defined by the appended claims.

Claims

1. A touch screen comprising: a display having an active area; an optical device in the active area at a position corresponding to a first region; a first row touch electrode disposed over the active area of the display and formed from a plurality of first touch electrode segments interconnected by a first plurality of bridges, wherein two segments of the plurality of first touch electrode segments separated by one or fewer first touch electrode segments are separated by a first distance; and a second row touch electrode formed from a plurality of second touch electrode segments, wherein two non-consecutive segments of the plurality of second touch electrode segments are separated by a second distance, less than the first distance.
